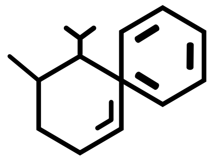
Doç. Dr. Murat Oluş ÖZBEK

GEBZE TECHNICAL UNIVERSITY

MUH - 391

HOLISTIC ENERGY OPTIMIZATION IN CAMPUS BUILDINGS



CREW

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DATE
26.03.2026

PROBLEM DEFINITION

- **University campuses have high energy consumption.**
- **A large portion of energy comes from HVAC (heating-cooling) systems.**
- **However, these systems often do not operate efficiently.**

Main problems:

1. Class schedules are used inefficiently
2. Empty spaces are unnecessarily conditioned
3. User behaviors cause energy waste

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PROJECT TOPIC

Developing a holistic system that optimizes energy consumption in campus buildings.

- **Industry:** optimization & decision system
- **Civil:** insulation solutions
- **Computer:** software & data
- **Chemistry:** energy modeling

Approach:

- Insulation + HVAC + occupancy density are considered together
- An interdisciplinary solution is applied

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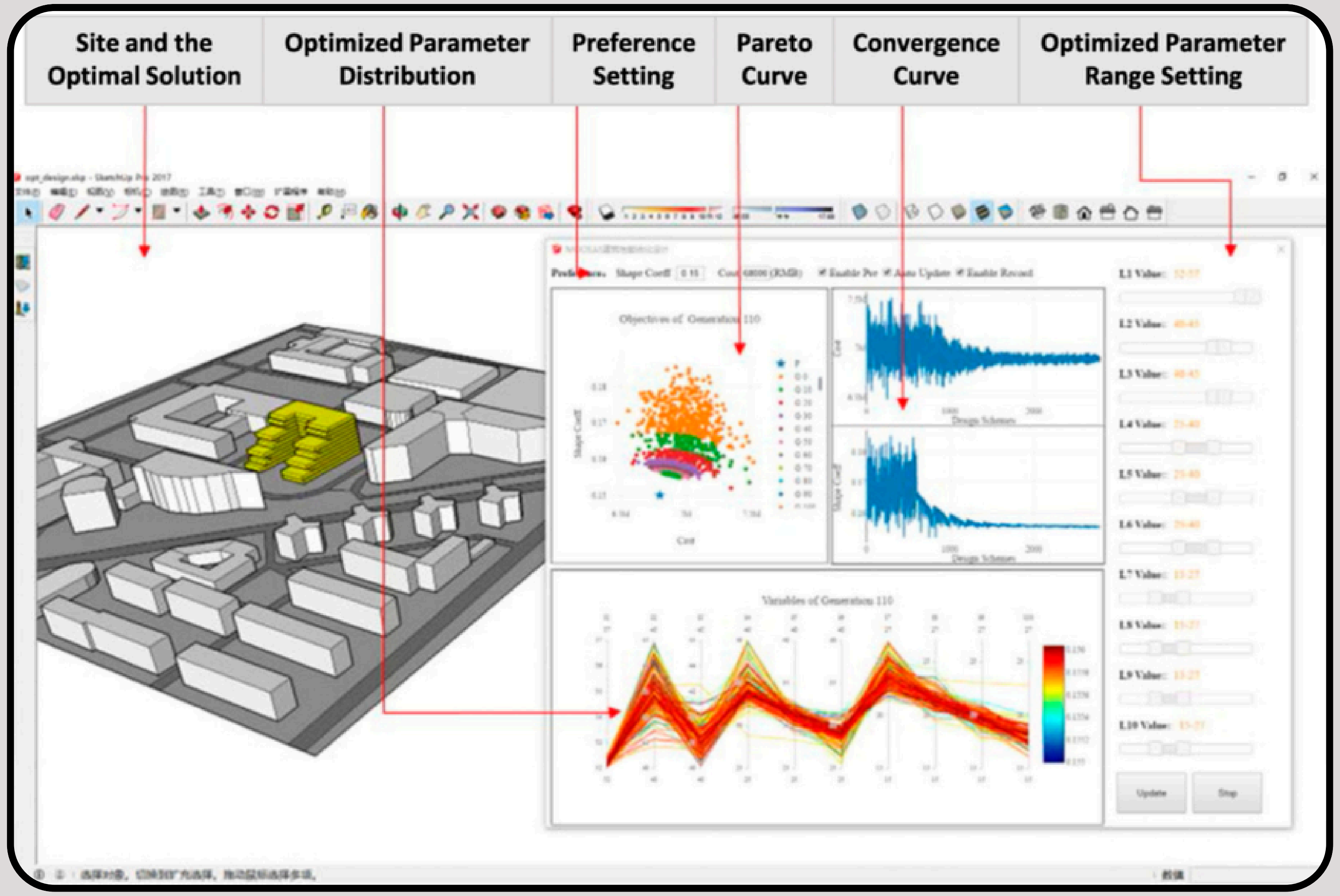
PROJECT OBJECTIVE

Developing an integrated model that reduces energy waste.

Targets:

- Smart air conditioning according to class schedule
- Reducing energy demand with insulation
- Establishing a simulation + optimization system
- Including user behavior in the model
- Developing an applicable and user-friendly tool



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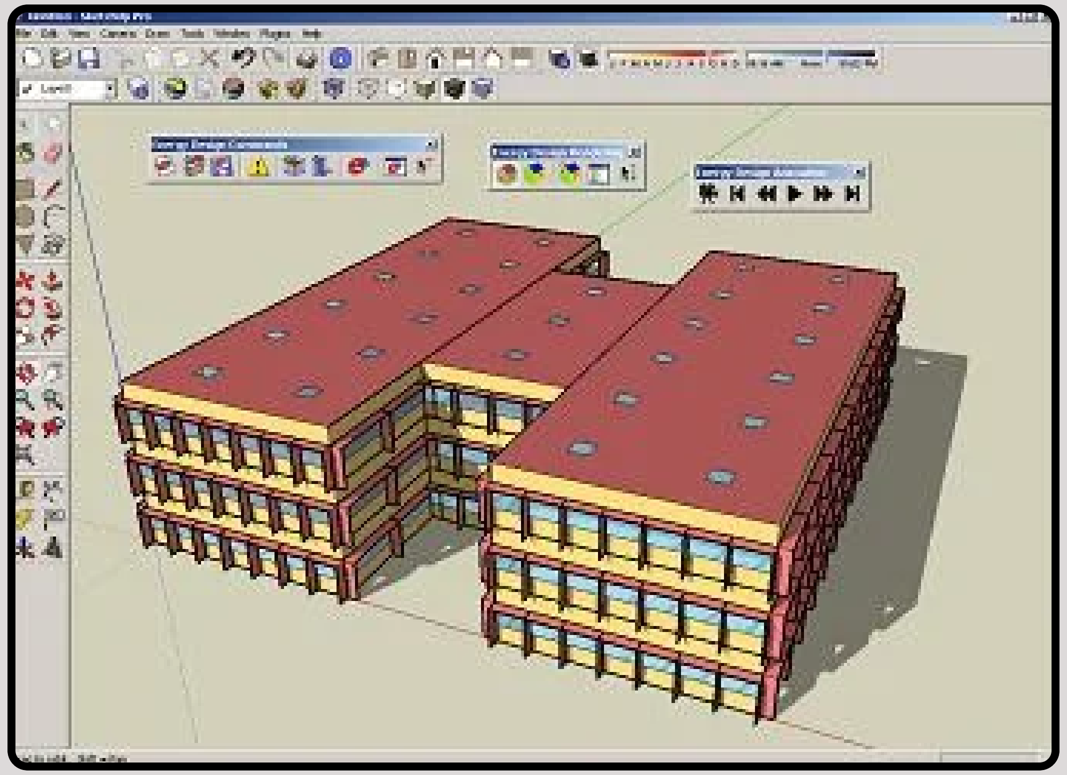
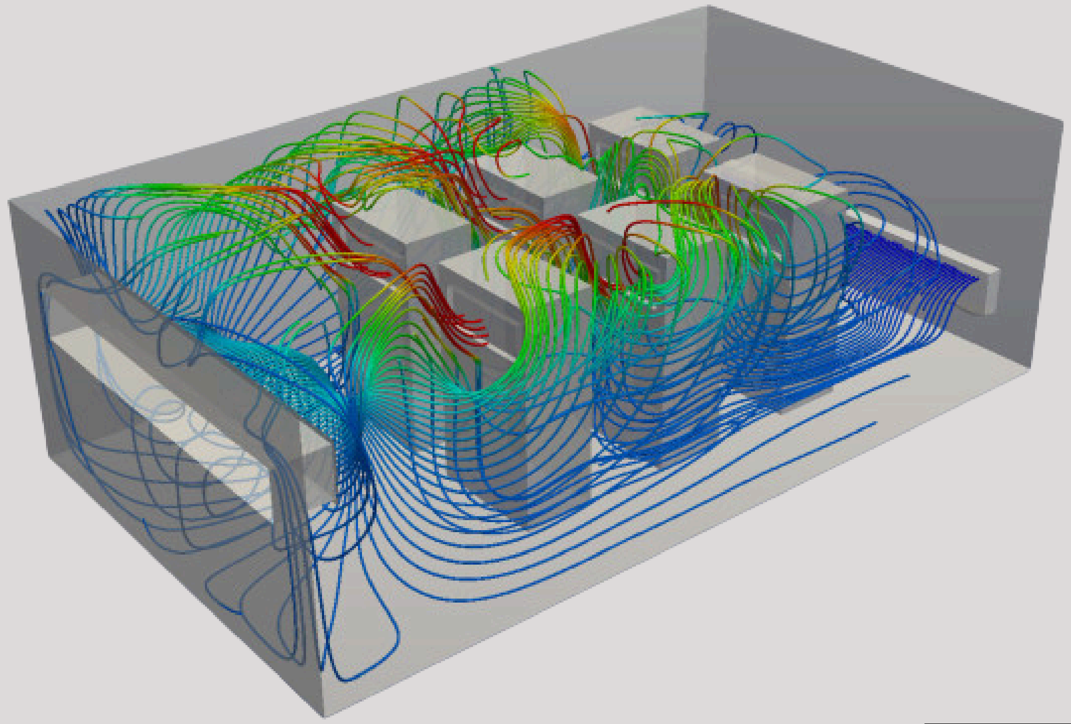
PROJECT SCOPE



What will be done in the project:

- Collection of building data
- Creating an energy simulation model
- Adding occupancy density to the model
- Insulation and structural improvements
- Analysis of HVAC scenarios
- Economic analysis (cost & payback)



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Teşekkürler

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